



Safety and Security in Digital Finance

Ass. Prof. Christina Kolb



I am...

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Faculty: Behavioural, Management and Social Sciences (BMS)

Group: FEBIS Financial Engineering and Business Information Systems

Research

- safety and security modelling and assessment

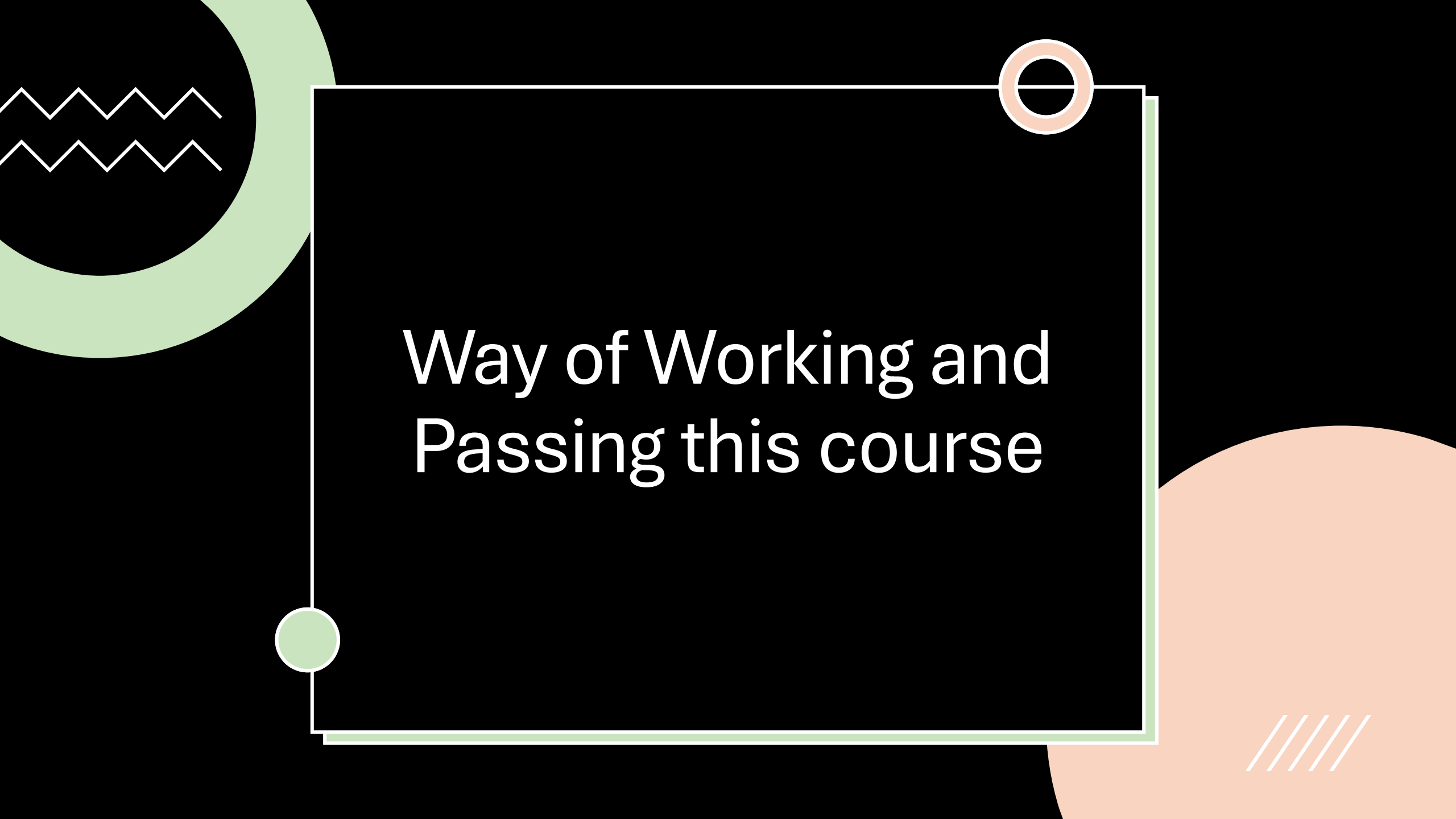
Teaching

- Master Course for BIT Enterprise Security Management: Security Experiments 1+2
- Master Cyber risk management Course for CS
- Bachelor Module 8 BIT
- Professional Learning and Development (PLD) Cybercrime Cybersecurity Risk Management



A magnifying glass with a dark frame is positioned over a light brown, textured background. The lens of the magnifying glass is centered and contains the text "WHO ARE YOU?" in a bold, dark, sans-serif font. The text is arranged in two lines: "WHO ARE" on the top line and "YOU?" on the bottom line. The magnifying glass's handle extends from the bottom right corner of the frame.

**WHO ARE
YOU?**



Way of Working and Passing this course

You are responsible!

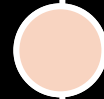
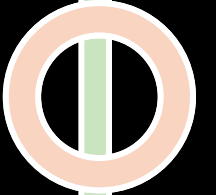
You take notes and add content to the slides!

You discuss with each other!

You read the slides and all content of this course!

You make sure to understand the content!

Use definitions and vocabulary!

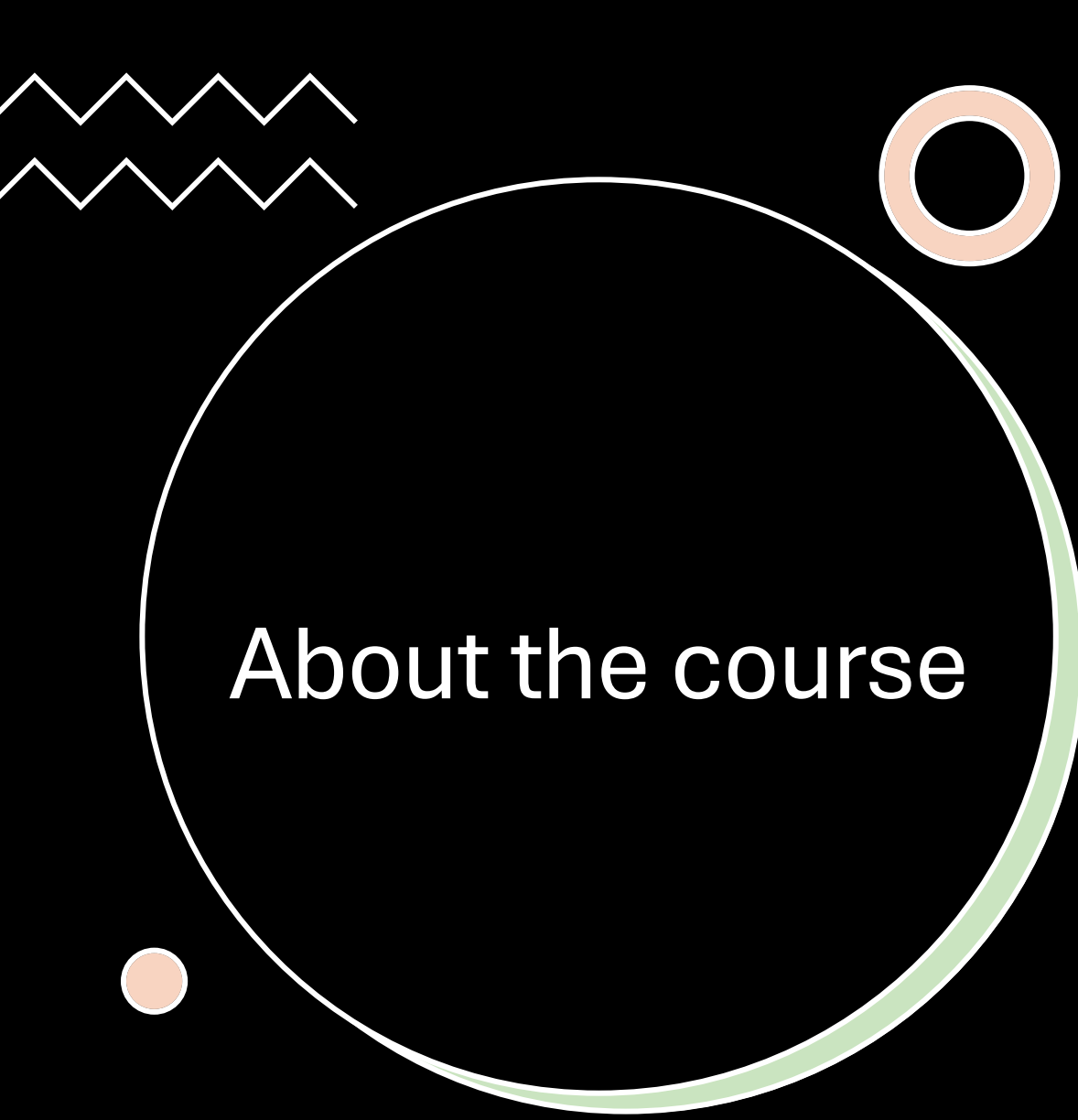


You are a
self-driven
learner!

Be PROACTIVE

HABIT #1





About the course

Cybersecurity in Digital Finance.

- Explain cybersecurity risks from social, software, hardware, organisational, and regulatory perspectives.
- Discuss how human behaviour and social engineering affect cybersecurity risks.
- Understand safety–security interactions in digital and financial systems.
- Analyse cybersecurity approaches to fraud detection and prevention.
- Connect cybersecurity concepts to their own doctoral research.





How to pass Cybersecurity in Digital Finance

3 ECTS

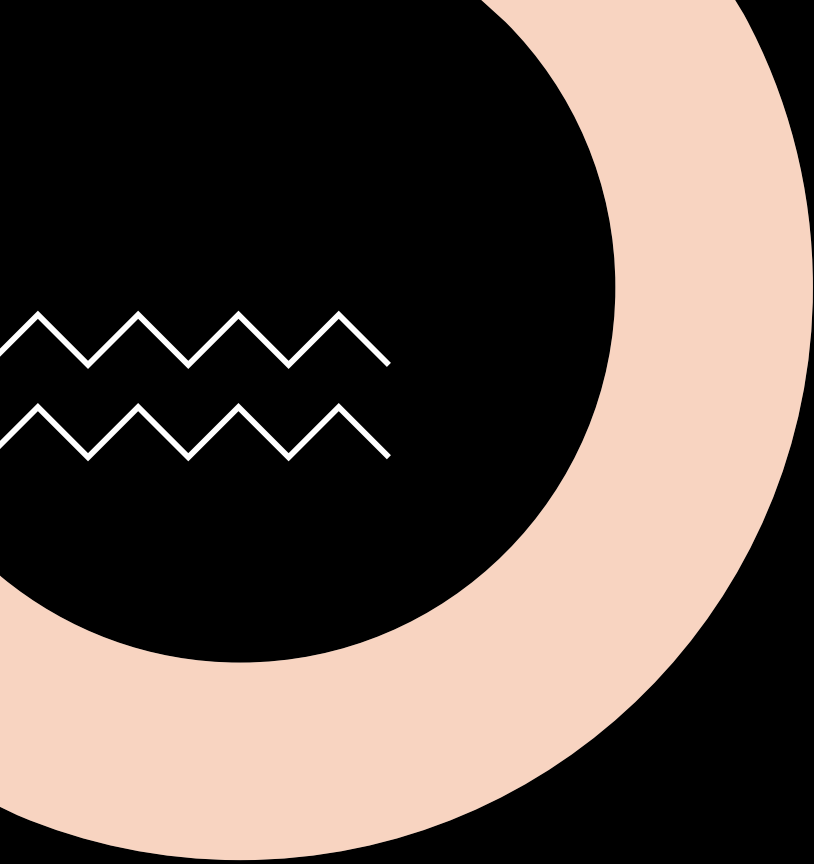
Participants are assessed on:

- Active participation in lectures, expert sessions, workshops, and supervised work.
- Engagement with cybersecurity topics and contribution to discussions.
- Preparation and delivery of a short presentation or report idea.
- Completion of a report connected to cybersecurity in digital finance.



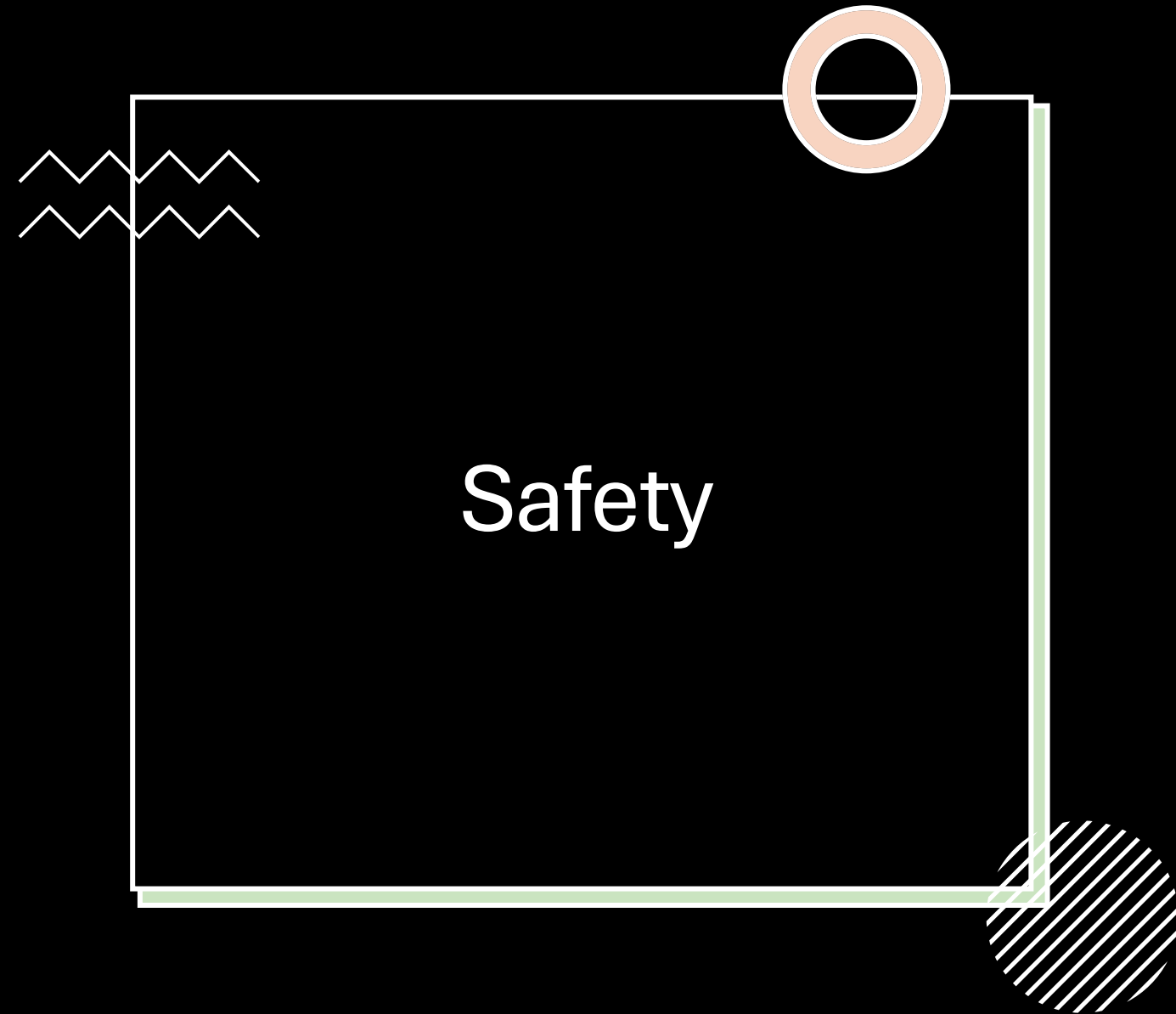
Important Dates:

Submit report: End of September



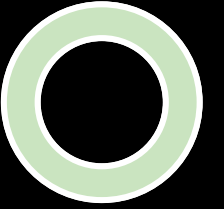
Safety



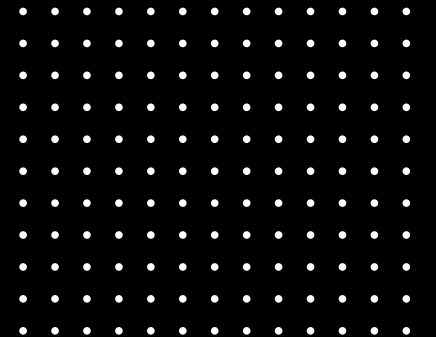


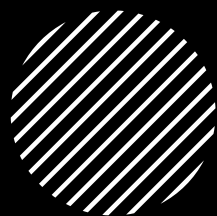
Safety refers

- to the absence of unintended failures
- or that we feel well as a person.



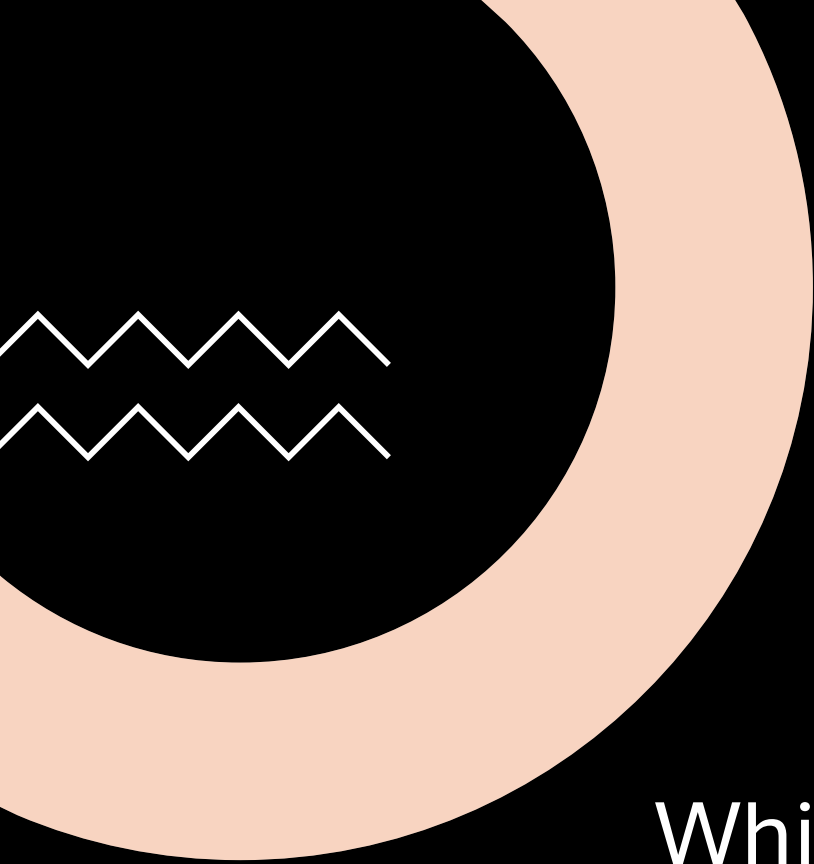
Which examples do you know for safety?





Which vocabulary is related to safety?





Which safety models do you know?

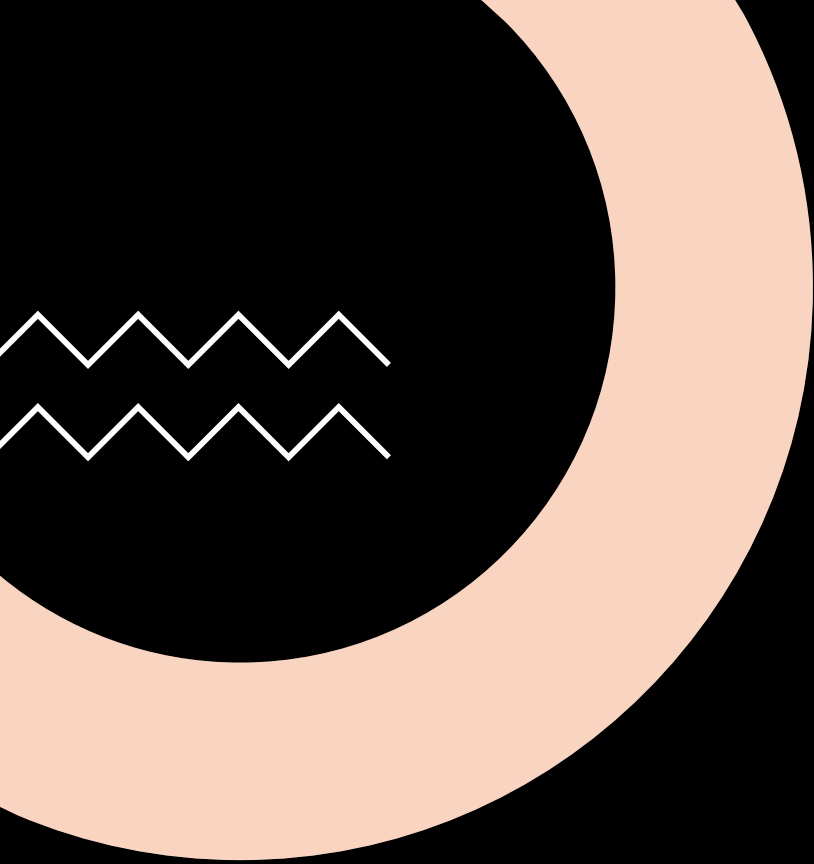




Which safety models do you know?

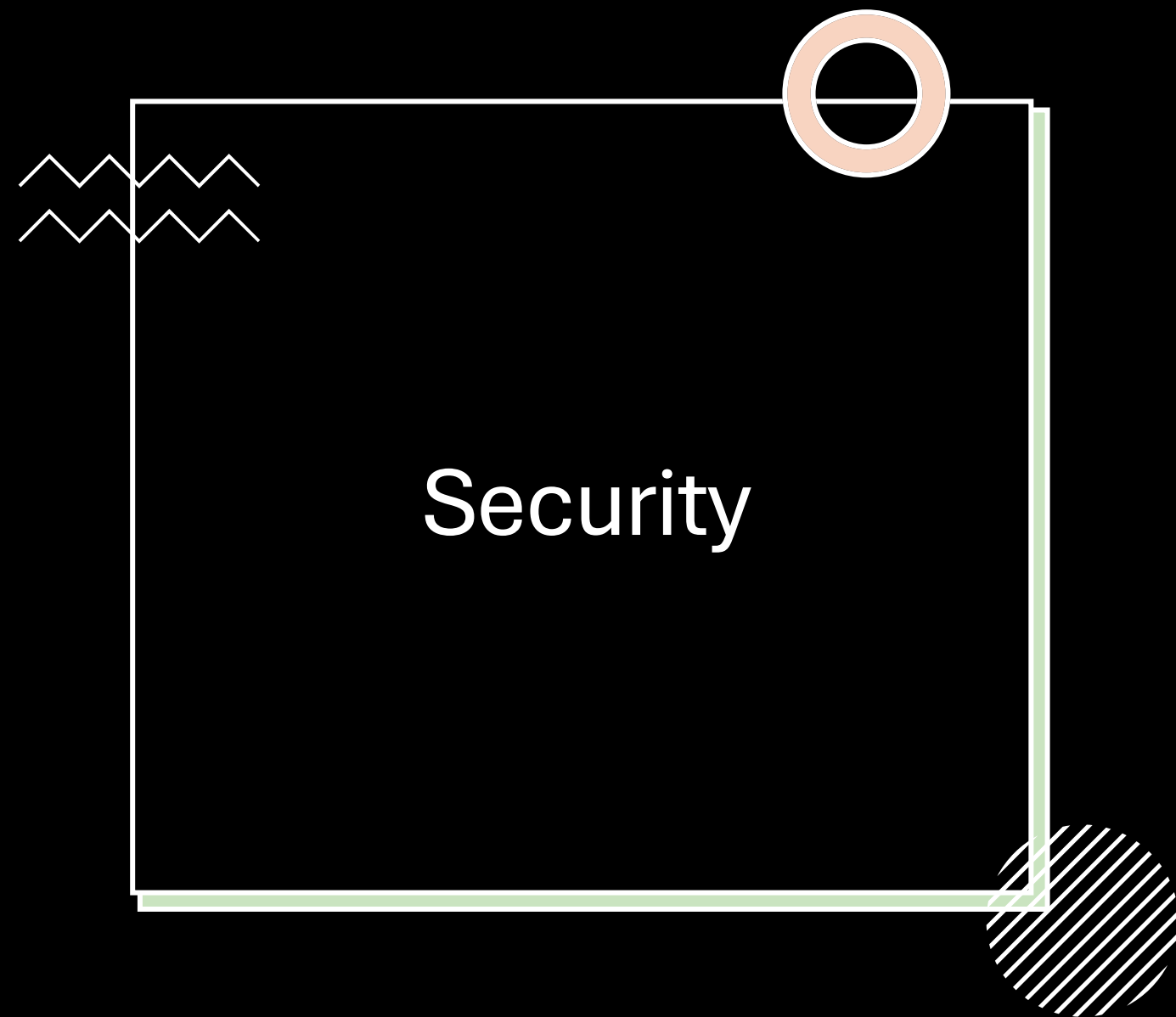
We will introduce fault trees soon.



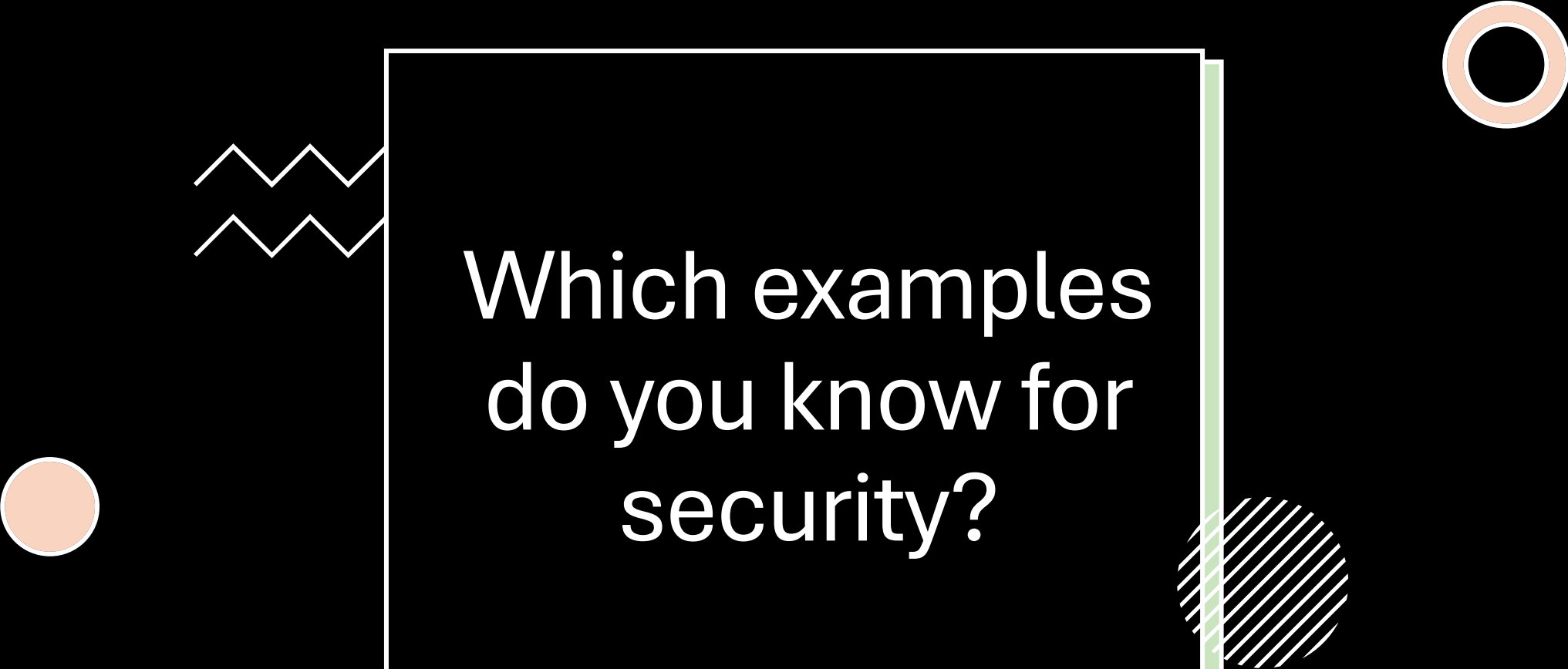


Security





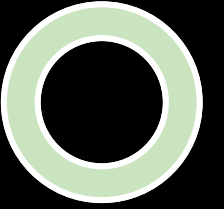
Security
refers to the absence
of malicious attacks.



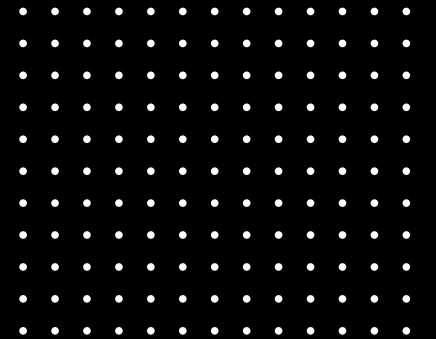
Which examples
do you know for
security?

Which vocabulary is
related to security?





Which security models do you know?





Why Safety and Security together?



Do you lock or unlock your door at night?



or



Open Door For Safety



No Security With Open Door



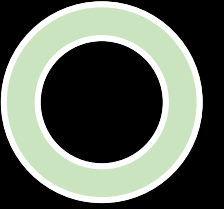
No Security With Open Door



This conflict between safety and security is called antagonism.



Attack Trees

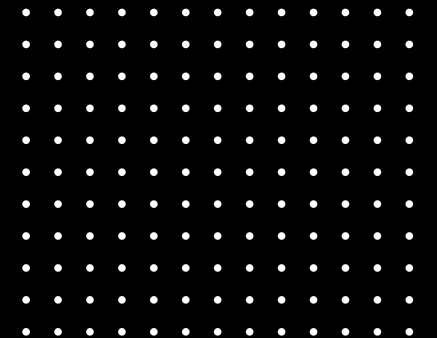


- Origin:
- Introduced in 1991 by Weiss
- To describe threats on computer systems and possible attacks

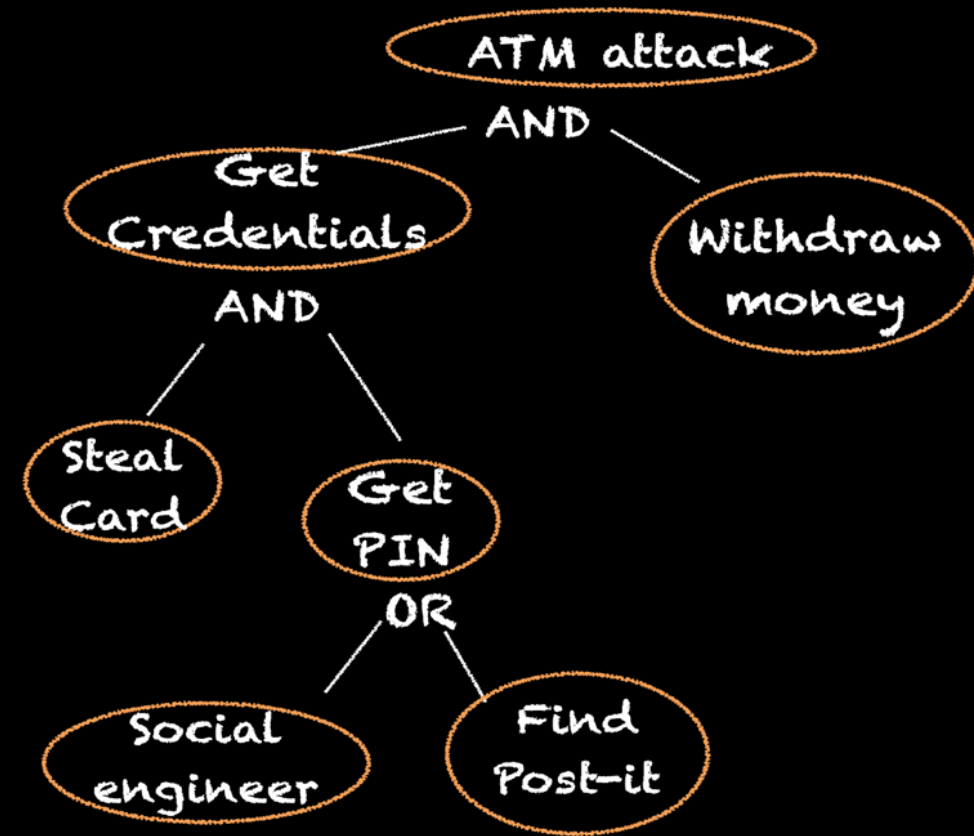
Structure:

- Basic events
- Intermediate Events
- Top Event
- And/ Or gates

Attack Trees

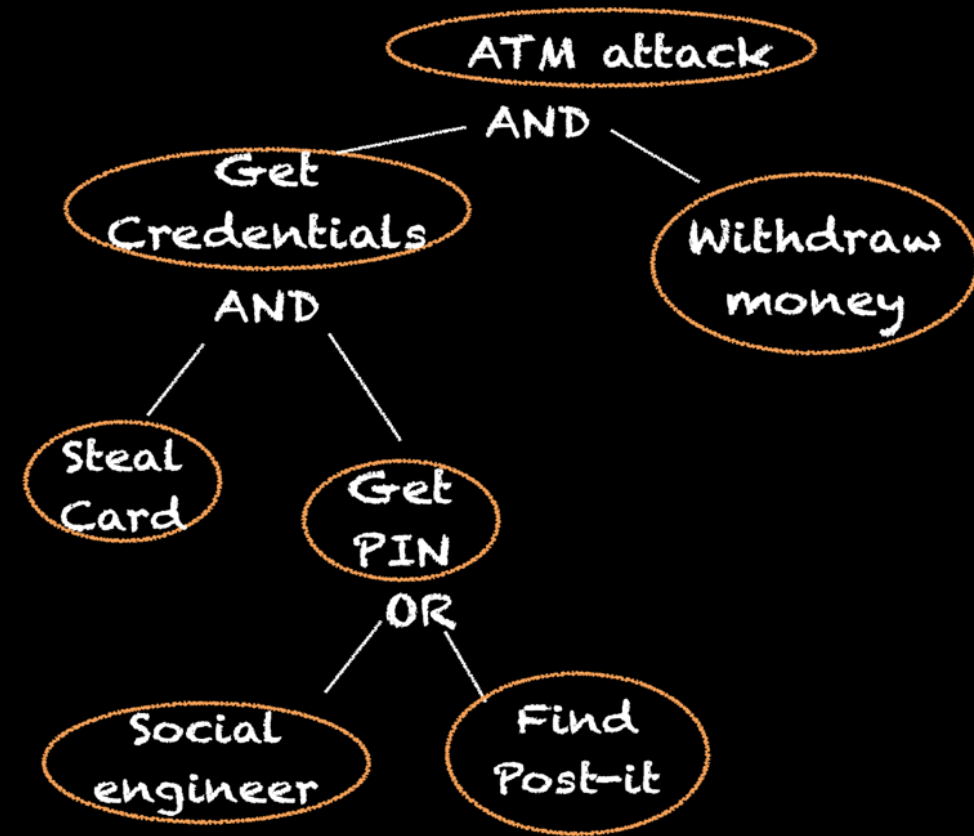


Attack Tree Example by Schneier



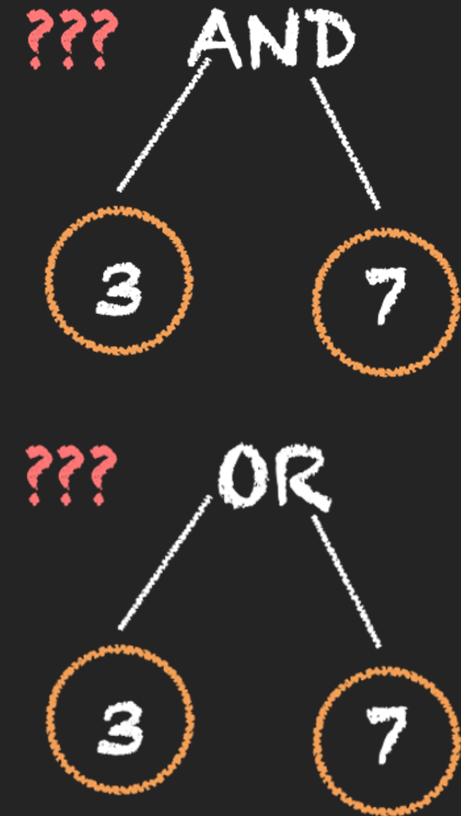
Attack Tree Example by Schneier

- Basic events?
- Top event?
- Intermediate events?



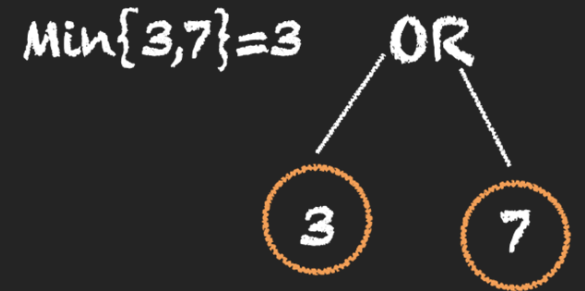
Propagation of Values Through Logical Gates

- Quantitative queries can be computed bottom-up.
- The values of the basic elements are propagated upwards in the tree
- Example:
Basic elements a and b cost resp. 3 EUR and 7 EUR



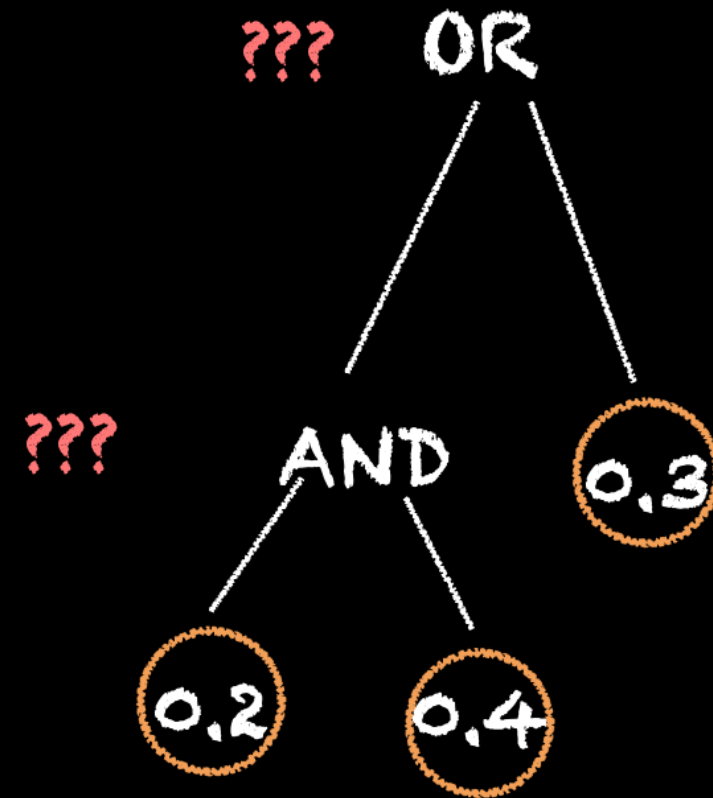
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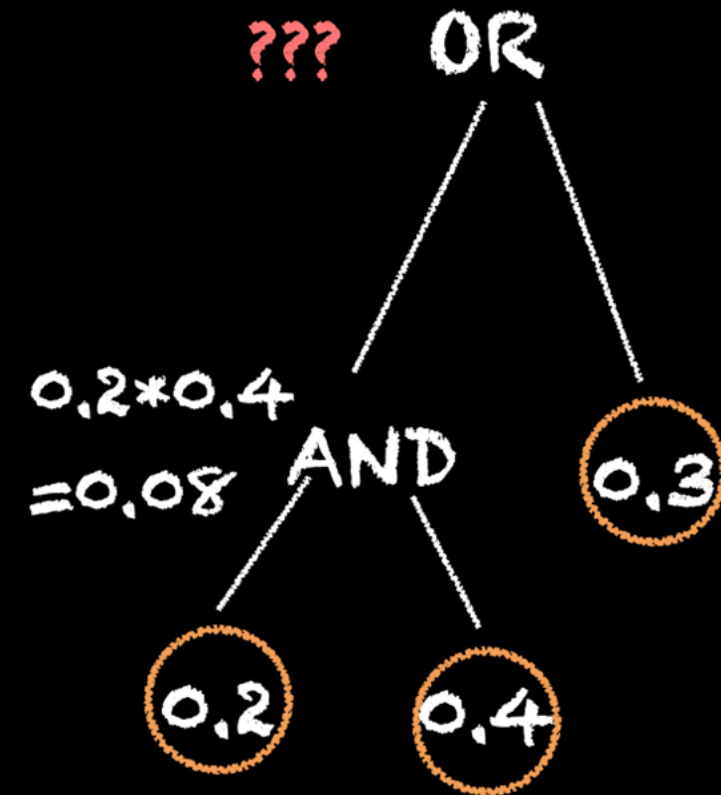
Max Probability

- The value of an or gate in an attack tree is the max value among its children.
- - The values in the basic elements are given
- The probability of an and gate is the product of its children



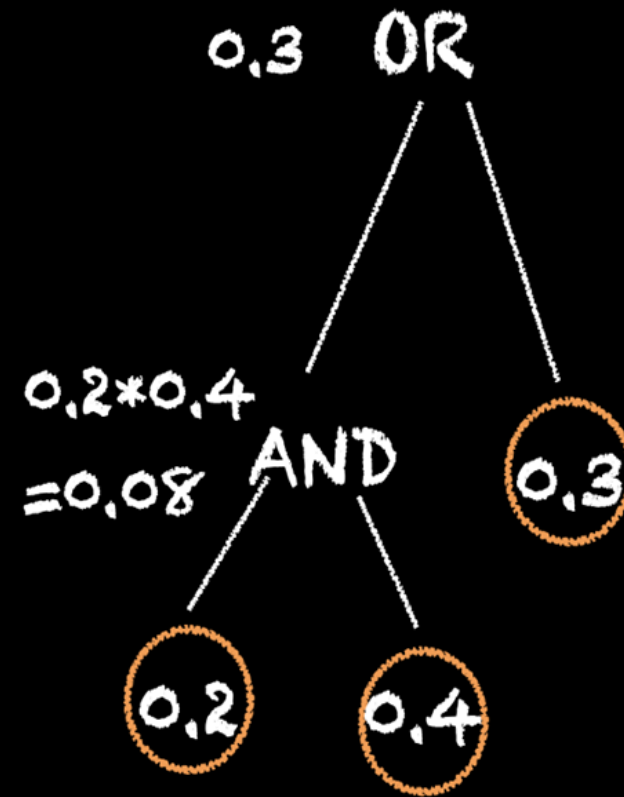
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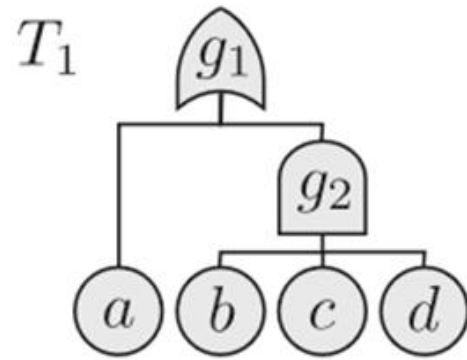
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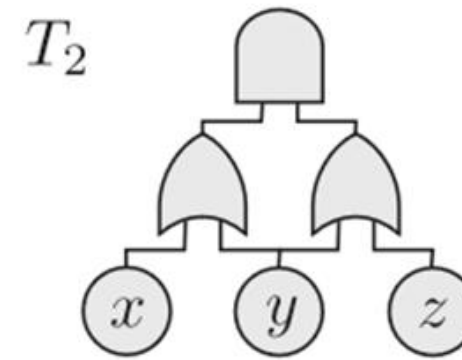
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(a) $\text{OR}(a, \text{AND}(b, c, d))$



(b) $\text{AND}(\text{OR}(x, y), \text{OR}(y, z))$

Fig. 1. T_1 has tree structure; the OR gate g_1 is the root; g_2 is an AND gate. T_2 has DAG structure: y has two parents.

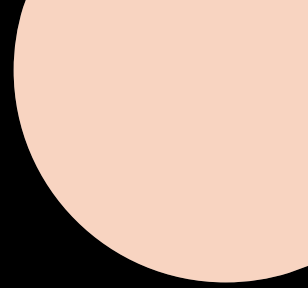
DAG Structure

- [QEST'21]: Attack Trees vs. Fault Trees: Two Sides of the Same Coin from Different Currencies

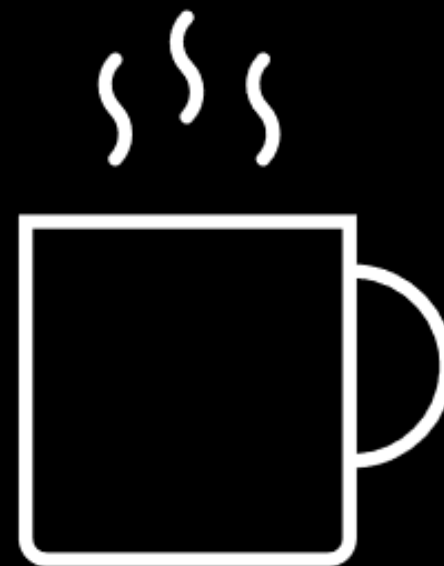
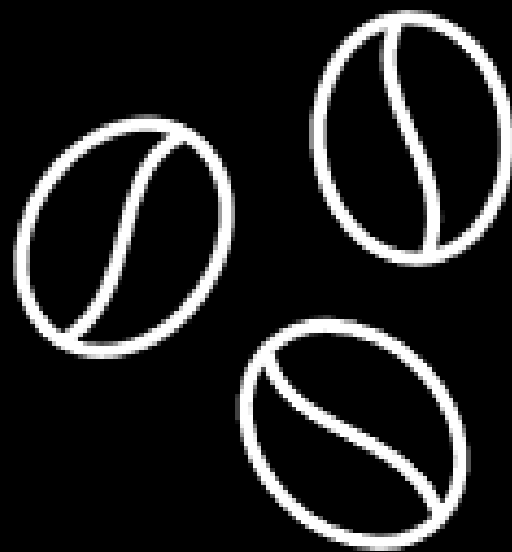


Search for extensions for attack trees and present them!

Which extensions do you find?



Break



Tutorial

- Model a case study of your choice with an attack tree
- Discuss your attack tree with one course participant
- Present attack tree models

